

A SIMPLE DIRECT METHOD FOR FAST TRANSIENT STABILITY ASSESSMENT OF LARGE POWER SYSTEMS

Y. Xue (*)

Th. Van Cutsem

M. Ribbens-Pavella

Department of Electrical Engineering; University of Liège
Sart-Tilman, B - 4000 Liège, Belgium

Abstract. A method for on-line transient stability assessment of large power systems is proposed. It consists in 1) replacing the multimachine system by a two-machine dynamic equivalent further amenable to a one-machine-infinite-bus system; 2) reducing the stability problem to a sole algebraic equation, devised from the well-known equal area criterion, or equivalently from the Liapunov direct criterion; 3) using this equation to derive one-shot stability analysis strategies. Moreover, a technique for system admittance matrix reduction is developed, which proves particularly efficient, especially for large systems and multiple contingency evaluation. The method's main appeal is rapidity: compared to the most efficient direct criterion, it is found to be about one order of magnitude faster. Other attractive features are flexibility and ability to encompass various simulation conditions. Extensions to on-line sensitivity analysis and control are also suggested.

1. INTRODUCTION

In a broad sense, developments in dynamic security monitoring and control of electric power systems are far behind static counterparts. In the specific area of transient stability assessment, significant achievements have been accomplished through Liapunov's direct criterion [1,2,3]. Yet, further improvements are still needed for effective on-line contingency evaluation strategies.

The most efficient Liapunov functions available nowadays belong to the class of energy type. Two subclasses may mainly be distinguished, presenting specific pros and cons [3,4]. (i) The "standard" functions are based on the "reduced" transmission network model, offer very interesting computational possibilities but impose neglect of transfer conductances. (ii) The "structure preserving" Liapunov functions are based on an "unreduced" transmission network model; they allow to take into account the main part of the transfer conductances, prove more flexible for better load and machine modeling, but they are computationally more demanding.

The method proposed in this paper attempts to combine and amplify the advantages of the above Liapunov approaches while circumventing some of their major difficulties. The leading idea is to reduce the multimachine problem to a simple equivalent and to apply available direct methods further modified so as to eventually get simple algebraic expressions.

The multimachine system is decomposed into a "candidate critical" machine and the remaining machines, aggregated to an equivalent one. The former is a machine (or cluster of machines) likely to be responsible of the system's separation, should the simulated disturbance be sustained. It is identified via the accelerations criterion [3] used to "pre-filter" candidate critical machines;

these are then tested in a sequence, in order to identify the actually critical one. On the other hand, the aggregation uses the center of angles concept [5,6]. The aggregation rule amounts to "freezing" the rotor angles of the machines belonging to the same group, and to replacing their time evolution by a common center of angles trajectory. Note that all electrical connections and all machines' electromotive forces are kept individually; transfer conductances are not neglected. The obtained two-machine equivalent is further reduced to a one-machine-infinite-bus system.

Using the above dynamic equivalents together with the well-known equal area criterion, combining them further with Taylor series expansions and corrective factors, we devise a simple analytic direct methodology. Its advantages are manifold. First, it allows performing one-shot stability analysis free from any step-by-step or trial and error procedures; besides, this analysis is applicable to any type and location of disturbances, and to multiple sequences of events with moreover different pre-fault and post-fault system configurations. Second, the method provides stability margins, expressible analytically. These in turn open possibilities towards on-line analytic sensitivity analysis and preventive control. Other applications could also be thought of, capable of meeting specific requirements for both planning and on-line purposes. Nevertheless, the emphasis in this paper is put more on the derivation of the very method and exploration of its performances.

We would like to mention that the idea of using dynamic equivalents has been investigated independently by Rahimi and Schaffer [7] and by ourselves [8]. However there exist significant differences between the two approaches. For example, as recognized by the authors of Ref. [7], their stability criterion allows the assessment only of the "worst" cases, i.e. those corresponding to three-phase short circuits applied at generator buses, when, furthermore, the short-circuited generator coincides with the critical one; besides, the stability assessment is computed in a trial and error fashion. Other important differences between the two approaches are indicated in § 3.6 below.

In all direct methods, a good deal of the computational effort bears on the reduction of the system admittance matrix at the generator nodes. In the present case, where the derived method is particularly little demanding, this reduction becomes a truly crucial factor. We have thus been led to develop a general technique for meeting this problem. This technique proves very efficient within the context of multiple sequential contingency simulation. Other method's figures are quoted in Section 5 where it is described.

2. DEVISING SIMPLE EQUIVALENTS

2.1. The multimachine model

In the classical simplified model of an n-machine power system, each generator is represented electrically as a constant voltage source behind its transient reactance and each load as a constant impedance. The motion of the i-th machine is accordingly described by

$$\delta_i = \omega_i ; \quad M_i \dot{\omega}_i = P_{mi} - P_{ei} , \quad i=1, \dots, n \quad (2.1)$$

where

$$P_{ei} = E_i^2 Y_{ii} \cos \theta_{ii} + \sum_{j=1, j \neq i}^n E_i E_j Y_{ij} \cos(\delta_i - \delta_j - \theta_{ij}) \quad (2.2)$$

(*) On leave from Nanjing Automation Research Inst., China.

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δ_i rotor angle
 ω_i rotor speed
 M_i inertia coefficient
 P_{mi} mechanical input power
 P_{ei} electrical output power
 E_i voltage behind the direct-axis transient reactance
 Y admittance matrix reduced at the internal generator nodes

Y_{ij}, θ_{ij} modulus and argument of the ij -th element of Y
 and where traditionally M_i, P_{mi}, E_i are assumed to be constant throughout the transients.

2.2. The equivalent two-machine model

Let us consider the relative motion of a specific machine with respect to the remaining system. Presuming that this machine moves apart from the others, we replace these latter by an equivalent, aggregated to one, machine.

Let us denote by
 s the specific machine; in the sequel we will refer to as the "candidate critical" machine (*);
 A the set of all remaining $(n-1)$ machines;
 a its equivalent, aggregated to one, machine.

In order to model the equivalent machine (a) and its motion, we will use the standard "center of angles" (COA) concept, while however considering only the $(n-1)$ machines of A [7,8]. Indeed, unlike to a current practice, it is physically more sound to exclude machine s from COA. We get accordingly

$$M_a = \sum_{l \in A} M_l ; \delta_a = M_a^{-1} \sum_{l \in A} M_l \delta_l ; \omega_a = \dot{\delta}_a ; \gamma_a = \dot{\omega}_a = \ddot{\delta}_a . \quad (2.3)$$

The motion of A is now readily obtained by summing up the corresponding $(n-1)$ eqs. (2.1), or equivalently eqs. (2.4);

$$M_l \ddot{\delta}_l = P_{ml} - P_{el} \quad \forall l \in A , \quad (2.4)$$

and by using relationships (2.3). We get

$$M_a \ddot{\delta}_a = \sum_{l \in A} (P_{ml} - P_{el}) . \quad (2.5)$$

In the above expressions (2.4), (2.5) P_{el} is of form (2.2). An additional simplification may be obtained by setting

$$\delta_j = \delta_a \quad \forall j \in A , \quad (2.6)$$

i.e. $\delta_s - \delta_l = \delta_s - \delta_a ; \delta_j - \delta_l = 0 \quad \forall l, j \in A . \quad (2.7)$

In this case, P_{el} expresses by

$$P_{el} = E_s^2 Y_{ll} \cos \theta_{ll} + E_l E_s Y_{ls} \cos (\delta_a - \delta_s - \theta_{ls}) + \sum_{\substack{j \in A \\ j \neq l}} E_l E_j Y_{lj} \cos \theta_{lj} \quad (2.8)$$

The above assumption (2.6) is physically sound and well validated by practice.

The motion of machine s is modelled by

$$M_s \ddot{\delta}_s = P_{ms} - P_{es} . \quad (2.9)$$

Here, P_{es} is expressed by an equation similar to (2.2), where in addition assumption (2.7) has been taken into account;

$$P_{es} = E_s^2 Y_{ss} \cos \theta_{ss} + \sum_{j \in A} E_s E_j Y_{sj} \cos (\delta_s - \delta_a - \theta_{sj}) . \quad (2.10)$$

2.3. The equivalent one-machine-infinite bus (OMIB) model

To derive the OMIB model, we consider the relative rotor angle and corresponding relative rotor acceleration

$$\delta \triangleq \delta_s - \delta_a ; \dot{\delta} = \dot{\delta}_s - \dot{\delta}_a . \quad (2.11)$$

Substituting eqs. (2.5), (2.9) into (2.11) we get

$$M \ddot{\delta} = P_m - P_e \quad (2.12)$$

(*) To simplify, we consider for the time being a simple critical machine. The generalization to a cluster of critical machines is discussed in § 3.3.

where

$$M = M_s M_a M_T^{-1} ; M_T = \sum_{i=1}^n M_i \quad (2.13)$$

$$P_m = (M_a P_{ms} - M_s \sum_{l \in A} P_{ml}) M_T^{-1} ; P_e = (M_a P_{es} - M_s \sum_{l \in A} P_{el}) M_T^{-1}$$

Using eq. (2.13) along with (2.8), (2.10) allows transforming eq. (2.12) in a more familiar expression of the equivalent OMIB equation of motion:

$$M \ddot{\delta} = P_m - [P_c + P_{\max} \sin (\delta - \nu)] = P_m - P_e \quad (2.14)$$

where

$$P_c = [M_a E_s^2 G_{ss} - M_s (\sum_{l \in A} E_l^2 G_{ll} + \sum_{l, j \in A; j \neq l} E_l E_j G_{lj})] M_T^{-1}$$

$$P_{\max} = [\sum_{l \in A} E_s E_l Y_{sl} (M_s^2 + M_a^2 - 2 M_s M_a \cos 2 \theta_{sl})^{1/2}] M_T^{-1}$$

$$\nu = \tan^{-1} \left(\frac{M_T}{M_a - M_s} \tan \theta_{s,a} \right) - \frac{\pi}{2} , \quad (2.15)$$

$\theta_{s,a}$ is the argument of the equivalent admittance between s and a :

$$\tan \theta_{s,a} = \left(\sum_{l \in A} B_{sl} \right) \left(\sum_{l \in A} G_{sl} \right)^{-1} , \quad (2.16)$$

and where B (G) stands for the susceptance (respective-ly the conductance):

$$G_{sl} = Y_{sl} \cos \theta_{sl} ; B_{sl} = Y_{sl} \sin \theta_{sl} . \quad (2.17)$$

3. THE PROPOSED METHOD

In this section, the OMIB formulation is used in conjunction with the equal area criterion and further developed to derive means for fast transient stability assessment. The method relies on the proper identification of the critical machine. This latter question is answered in § 3.3.

3.1. The equal area criterion (EAC) revisited

Changing the system configuration amounts to changing its matrix Y , and hence to changing parameters P_c , $P_{\max}$ and ν expressed by (2.15).

Plotting the P - δ curves provided by (2.14) and exploring them for the pre-fault or original (O), during fault (D), and post-fault (P) configurations is the basis of the well-known EAC [9], illustrated in Fig. 1. The original (steady-state) operation is characterized by the rotor angle δ_0 located at the crossing of the horizontal line $P = P_m$ with curve 0, partially drawn, and sketching the original P_{e0} curve. The post-fault

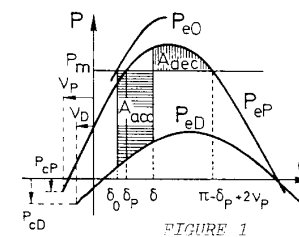


FIGURE 1

stable (respectively unstable) equilibrium point is determined by the intersection of P_m with P_{eP} ; this provides δ_p (respectively $(\pi - \delta_p + 2\nu_p)$). Moreover, the value that the δ angle reaches at the fault clearing time delimits the accelerating, A_{acc} , and decelerating, A_{dec} , areas, which measure the transient accelerating and decelerating energies.

The EAC simply recognizes the fact that the faulted system keeps capable of recovering stability as long as A_{acc} is smaller than A_{dec} . The borderline case corresponds to

$$A_{dec} = A_{acc} \quad (3.1)$$

for which the system reaches the critical clearing angle (CCA or δ_c). The conventional critical clearing time (CCT or t_c) is the time required by the faulted system to move from δ_0 ($t = t_0$) up to δ_c ($t = t_c$). Conversely, one may assess the system's robustness at a given clearing time in terms of the difference

$$A_{dec} - A_{acc} .$$

3.2. A fast CCT computation procedure

An analytical straightforward relationship between

a rotor angle and the corresponding (fault clearing) time, i.e. the time required to reach it, is the Taylor series expansion

$$\delta_r = \delta_0 + \frac{1}{2} \gamma \tau^2 + \frac{1}{24} \gamma^{(2)} \tau^4 + \frac{1}{720} \gamma^{(4)} \tau^6 + \dots \quad (3.2)$$

where δ_0 is the original steady-state rotor angle, and the successive derivatives $\gamma^{(2)}, \gamma^{(4)}, \dots$, are obtained at t_0^+ , i.e. immediately after the fault inception; notice that for continuity reasons, $\delta(t_0^+) = \delta(t_0^-)$ and that the odd derivatives of δ are zero at t_0^+ [10].

To get easy analytic expression of τ in terms of δ we will truncate the above expression after the third RHS term. To avoid deteriorating the accuracy when the contribution of the higher order terms is not negligible, we will introduce corrective factors.

(i) *Deriving the CCA expression.* By definition, δ_c is the angle verifying (3.1). Assuming that the angle δ indicated in Fig.1 represents δ_c and that

$$P_{eD} = P_{CD} + P_{\max D} \sin(\delta - \nu_D); \quad P_{eP} = P_{CP} + P_{\max P} \sin(\delta - \nu_P) \quad (3.3)$$

we get :

$$A_{acc} = (P_m - P_{CD})(\delta_c - \delta_0) + P_{\max D} [\cos(\delta_c - \nu_D) - \cos(\delta_0 - \nu_D)]$$

$$A_{dec} = (P_{CP} - P_m)(\pi - \delta_c - \delta_P + 2\nu_P) + P_{\max P} [\cos(\delta_c - \nu_P) + \cos(\delta_P - \nu_P)] \quad (3.4)$$

Equating the RHS members of the above expressions we get

$$\delta_c = \cos^{-1} \frac{R}{(A^2 + B^2)^{1/2}} + \cos^{-1} \frac{A}{(A^2 + B^2)^{1/2}} \quad (3.5)$$

where

$$A = P_{\max P} \cos \nu_P - P_{\max D} \cos \nu_D$$

$$B = P_{\max P} \sin \nu_P - P_{\max D} \sin \nu_D$$

$$R = P_m(\pi - \delta_0 - \delta_P + 2\nu_P) - P_{CD}(\delta_c - \delta_0) - P_{CP}(\pi - \delta_c - \delta_P + 2\nu_P) - P_{\max D} \cos(\delta_0 - \nu_D) - P_{\max P} \cos(\delta_P - \nu_P) \quad (3.6)$$

Notice that (3.5) is an implicit expression of δ_c (since R depends on δ_c) and must be solved iteratively. Experience based on a very large number of cases suggests that taking as initial value $\delta_c^0 = 0.6\pi$ contributes to making the iterative procedure extremely fast and to obtaining the solution in three iterations at the maximum.

(ii) *Expressing $\tau = \tau(\delta)$ for small τ values.* Solving for τ the Taylor series (3.2), truncated after the third RHS term, yields

$$\tau = \left[\frac{-b \pm (b^2 - 4dc)^{1/2}}{2d} \right]^{1/2} \quad (3.7)$$

where

$$d = \gamma^{(2)} = \gamma_s^{(2)} - \gamma_a^{(2)}; \quad b = 12\gamma; \quad c = -24[\delta_r - \delta_0] \quad (3.8)$$

(iii) *Corrective factors for longer clearing times.* Eq. (3.7) provides good results only when the contribution of terms higher than $0.5\gamma\tau^2$ are negligible. Two such cases may be distinguished :

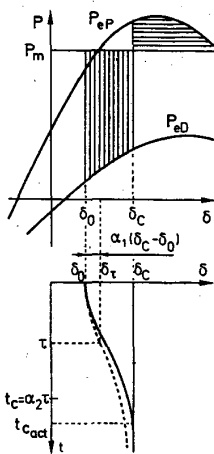


FIGURE 2

(1) the particular but practically interesting case where the considered disturbance is a three-phase short circuit located at the critical machine's busbar : γ_s is then a constant, $\gamma_s = P_{ms} M_s^{-1}$, and the higher than $0.5\gamma\tau^2$ order terms of the Taylor expansion vanish;

(2) τ is small enough to counterbalance the effects of the successive γ derivatives' terms; let us mention that in practice these derivatives may assume large values.

Let us examine the latter case first. Expression (3.2) shows that the factors multiplying $\gamma^{(4)}, \gamma^{(6)}, \dots$, increase much faster than linearly with time : the larger the τ , the more important the effects of the neglected terms. Fig.2 illustrates

the influence of $\gamma^{(4)}$ term on the computation of t_c and suggests the idea of using the Taylor series during a time τ smaller than t_c , then extrapolating for larger values. More specifically we propose to :

- compute δ_c via (3.5);
- introduce an angle factor α_1 , ($0 < \alpha_1 < 1$), defined by

$$\alpha_1 = (\delta_r - \delta_0) / (\delta_c - \delta_0), \quad 0 < \alpha_1 < 1 \quad (3.9)$$

and solve for δ_r ; obviously, $\delta_0 < \delta_r < \delta_c$;

- use eq.(3.7) to compute the corresponding τ ; obviously, $\tau < t_c$;
- introduce a time factor $\alpha_2 > 1$ and express the t_c corresponding to δ_c by

$$t_c = \alpha_2 \tau \quad (3.10)$$

To properly fix them, observe at first that these factors are not independent; rather, they must obey a relationship imposed by the limit case where all higher than τ^2 terms vanish. It is easily shown that the α_1, α_2 factors should then be related by

$$\alpha_1 \alpha_2^2 = 1, \quad (3.11)$$

since in this case the expression

$$\delta = \delta_0 + \frac{1}{2} \gamma \tau^2 \quad (3.12)$$

would be verified for any couple (δ, τ) and in particular for the couples $(\delta_c, t_c), (\delta_r, \tau)$ related by (3.9), (3.10). Indeed, using eqs.(3.10) to (3.12) we get successively

$$\delta_r - \delta_0 = \frac{1}{2} \gamma \tau^2 = \frac{1}{2} \gamma \frac{t_c^2}{\alpha_2^2} = \frac{1}{2} \gamma \frac{\alpha_1 \alpha_2^2}{\alpha_2^2} t_c^2 = \frac{1}{2} \gamma \alpha_1 \frac{2(\delta_c - \delta_0)}{\gamma}$$

i.e. we get condition (3.9).

Hence, in the limit case where (3.12) would be rigorously verified, we should fix a small α_1 value to comply with the observation of case (2) and compute α_2 by (3.11). Actually, we propose to fix $\alpha_1 \in [0.2, 0.4]$ and to use a learning simulations set to derive satisfactory α_2 value. In all simulations we have chosen $\alpha_1 = 0.3, \alpha_2 = 1.7$ (< 1.82 provided by (3.11)). A detailed error analysis and investigation concerning the choice of (α_1, α_2) is given in [11].

3.3. Identification of the critical machine

For a certain fault sequence occurring at either generator or non-generator buses, the (cluster of) critical machine(s) is identified as follows : (i) draw up a list of "candidate" critical machines; to this purpose, use the initial accelerations criterion to "pre-filter" the candidates efficiently; (ii) compute by turn the CCTs corresponding to each (cluster of) candidate(s); (iii) identify the (cluster of) critical machine(s) as the one giving the smallest CCT, which is considered to be the actual system CCT. The overall algorithm is sketched in [8].

The filtering capability of the accelerations criterion has proved very good. In practice, only two or seldom three machines having the largest initial accelerations

$$\gamma_i = [P_{mi} - P_{ei}(\delta(t_0^+))] M_i^{-1}$$

will be declared as the candidates. The computation of γ_i is extremely fast.

Comparing the candidate CCTs allows also providing good information about the critical coherent group. If several CCTs are close to each other and correspond to strongly connected machines (on the basis of Y matrix elements), we can set up a new candidate critical group and obtain the corresponding CCT.

Admittedly, the accelerations criterion used alone may exhibit non satisfactory machine separation for faults located elsewhere than generator buses or near a group of closely connected machines. Nevertheless it must be emphasized that this criterion does not intend to identify the critical machine; it only acts as a speed-up step of the proposed procedure. Incidentally, note that the critical machine identification depends more or less on other conditions such as the system final configuration or the fault sequence. These facts

are also taken into account in the candidate CCT computation and in the final critical machine identification.

3.4. Extensions

3.4.1. Successive actions during the fault-on period.

The way the method is modified to comply with a sequence of events rather than a single one is described in Fig. 3. A Taylor series is used for each event of the sequence; it requires computation of five parameters when truncated after the $\gamma^{(2)}$ term as previously, viz. $\delta, \omega, \gamma, \gamma^{(1)}, \gamma^{(2)}$. At the end of the i -th event, of duration Δt_i , these parameters are expressed in terms of those at the beginning of the same event (see in Ref. [7] our Discussion). For example,

$$\delta(t_i^+) = \delta(t_{i-1}^+) + \omega(t_{i-1}^+) \Delta t_i + \frac{1}{2} \gamma(t_{i-1}^+) \Delta t_i^2 + \frac{1}{6} \gamma^{(1)}(t_{i-1}^+) \Delta t_i^3 + \frac{1}{24} \gamma^{(2)}(t_{i-1}^+) \Delta t_i^4, \quad (3.13)$$

where $\delta(t^-)$ (respectively $\delta(t^+)$) denotes the value of δ at the instant immediately before (respectively after) t . Transition from t^- to t^+ obeys the boundary conditions $\delta(t_{i-1}^+) = \delta(t_{i-1}^-)$; $\omega(t_{i-1}^+) = \omega(t_{i-1}^-)$.

On the other hand, $\gamma(t_i^+), \gamma^{(1)}(t_i^+), \gamma^{(2)}(t_i^+)$ are computed from the new prevailing configuration.

Note that in the case of successive actions the stability assessment is carried out on the basis of the difference $A_{dec} - A_{acc}$, rather than CCT.

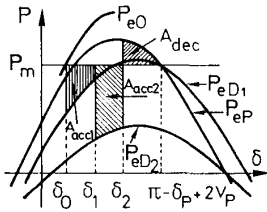


FIGURE 3

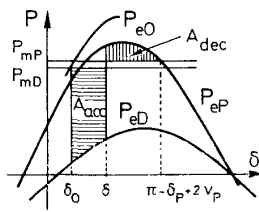


FIGURE 4

3.4.2. The post-fault mechanical power is different from the original one. This extension is interesting, because it allows considering generator shedding.

The P_m change affects the CCT through δ_p and $(\pi - \delta_p + 2\nu_p)$. Fig. 4 shows that it may be taken into account by merely changing the corresponding terms of expression (3.6) of R.

3.5. Transient stability assessment strategies

Various practical strategies and measures may be thought of. Two of them are briefly discussed hereafter.

3.5.1. The CCT. This is the conventional measure of a system's robustness. The procedures devised in §§ 3.2, 3.4 make its assessment extremely fast, flexible vis-à-vis various kinds of scenario, and reliable within the simplified model's validity limits.

3.5.2. The transient security margin, η . At a given clearing time τ the system's security margin may be assessed through

$$\eta_1 = \frac{A_{dec}(\tau) - A_{acc}(\tau)}{A_{acc}(\tau)} \quad \text{or} \quad \eta_2 = A_{dec}(\tau) - A_{acc}(\tau) \quad (3.14)$$

where $A_{dec}(\tau), A_{acc}(\tau)$ are obtained by expressions similar to (3.4), appropriately adapted to δ_τ , computable through the Taylor series (3.2). Notice that the computational efficiency and accuracy of this margin are even better than those of CCT since in practice τ assumes values generally smaller than CCT.

3.6. Method's keypoints and particular features

Let us recollect and comment some interesting aspects of the method developed in Sections 2 and 3.

(i) Excluding the critical machine from the aggregated COA model of the remaining system is physically sound [13]; according to our experience this has beneficial

effects on the accuracy.

(ii) Freezing all angles within the same group amounts to taking off the transient potential and kinetic energies that do not contribute to the separation in two subsets.

(iii) All static parameters included in Y , are kept individually.

(iv) Similarly, all E_i 's are kept individually.

(v) All transfer conductances are taken into account without making the procedure path-dependent as is the case in the multimachine Liapunov energy function.

(vi) Truncating the $\gamma^{(2)}$ term of the Taylor series would have provided much easier analytic expressions. However, experience shows that this may deteriorate significantly the accuracy. On the other hand, keeping the term in $\gamma^{(4)}$ would have cleared the method from its very rare inaccuracies; almost all discrepancies appear for large value of CCT (> 0.50 sec.) and $\gamma^{(4)}$ positive and larger than 2000, whereas its current values are less than $1000 \text{ rad. sec}^{-4}$. Nevertheless, this would have deprived the method from the essential advantage of providing algebraic, straightforward expressions.

(vii) The method is by essence extremely fast: it is a one-shot procedure, without intermediate computations in between δ_0 and τ . Hence, it is much faster than the fastest Liapunov direct method.

(viii) Incidentally, let us observe the conceptually close similarity of the two-machine Liapunov energy type function and the EAC methods [8]. Let us also observe that it is also possible to get a straightforward analytic expression with the multimachine Liapunov function, provided however that we neglect $\gamma^{(2)}$ term in the Taylor series; but the results thus obtained are reliable for CCTs not exceeding 0.45 sec. [3].

(ix) The method proves very flexible, able to treat sequence of events, any fault type and location, any type of system original and final configurations.

Let us quote some similarities and differences of our method with the related method of Ref. [7].

* **Similarities** : - they both use system dynamic equivalents in conjunction with EAC; moreover, - to construct the aggregated COA model of the remaining system A, they both exclude the critical machine (see above (i)), unlike other related approaches.

* **Differences** : In Ref. [7] : - the single-machine-infinite-bus equivalent is developed only for that generator where the three-phase-short circuit is presumably located; no cluster of critical machines is considered. Hence, the stability assessment concerns the "worst" cases only; - the CCT is computed in a trial and error fashion, the sequence of events always being of the type "application of three-phase-short-circuit-breaker opening - breaker reclosure" (see in [7] our Discussion); - no algebraic simple expression is provided, able to express stability margins and to allow easy sensitivity analysis.

4. SENSITIVITY ANALYSIS

The sensitivity analysis is a basic step for (i) offering more detailed information to operators, such as how much more change of a certain parameter can be suffered without losing security; (ii) supervising preventive control; (iii) performing optimal power flow with dynamic security constraints [14].

Generally, the on-line violation of dynamic security constraints leave very short time for operators to take corrective actions; thus, preventive analysis is more critical and important for on-line dynamic corrective control than static or off-line ones. Two kinds of approaches may be distinguished: numerical and analytical. The former is hardly feasible on-line. On the other hand, no analytical approaches are as yet available.

The simple formulation we developed in the previous

Sections may be extended and exploited for on-line analytical sensitivity analysis. The leading idea is as follows : using the same as above assumptions, we fix all $\delta_{\lambda} \forall \lambda \in A$ at δ_a . Then, for the machine of concern we compute the sensitivity of the corresponding η_2 (3.14) to control variables, such as the active power productions :

$$\frac{\partial}{\partial P_{ms}} \eta_2(P_{ms}, \delta_o, \delta_p, \gamma, \gamma^{(2)})$$

Notice that computation of the equivalent angle δ_p can be obtained directly from the sin curves of Fig.1. On the other hand, since $\delta_{\lambda} = \delta_a \forall \lambda \in A$, the formulation of $\gamma, \gamma^{(2)}$, and of their partial derivatives becomes simple.

Overall, the main computation effort required by the suggested sensitivity analysis is that of the original load flow, and of the Y matrix reduction for the during-fault and post-fault configurations. Since the corresponding matrices are already available, the analysis proves very straightforward.

5. Y MATRIX REDUCTION TECHNIQUES

The above developments show that the main computational complexity of the proposed method bears on the reduction of the involved admittance matrices. This section proposed a technique for calculating these matrices, especially appropriate for multiple contingency evaluation. Because of the repetitive nature of this computation, we concentrate on practical schemes which exploit common calculations and re-use intermediate results to the greatest extent.

5.1. Computing the original (pre-fault) matrix Y_o

$$\text{By definition } I_f = Y_o E_f \quad (5.1)$$

where Y_o is the $(n \times n)$ reduced admittance matrix seen from the machine internal nodes and subscript "f" refers to these fictitious nodes. Eq.(5.1) suggests a way to determine the elements of Y_o ; indeed :

$$[Y_o]_{ij} = I_i]_{E_k=0, k \neq j; E_j=1} \quad (i, j = 1, \dots, n) \quad (5.2)$$

Let us consider now the unreduced network with the loads replaced by shunt impedances and each machine by a current injector in parallel with the transient reactance (Fig.5). The "unreduced version" (superscript "u") of Eq.(5.1) is the $(N \times N)$ system :

$$\begin{bmatrix} I_m \\ 0 \end{bmatrix} = \begin{bmatrix} Y_{mm}^u & Y_{m1}^u \\ Y_{m1}^{uT} & Y_{11}^u \end{bmatrix} \begin{bmatrix} V_m \\ V_1 \end{bmatrix} \quad (5.3)$$

where the node set has been partitioned into machine and load buses. The idea of the proposed method is to obtain each column of the Y_o matrix as the solution of (5.3) with $E_k=0$ ($k \neq j$) and $E_j=1$, i.e. with (see Fig.5)

$$I_m = [0 \dots \frac{1}{jX_j} \dots 0]^T \quad (5.4)$$

To this end the Y^u matrix is first factorized (using optimal ordering) into $Y^u = LDL^T$ (5.5)

where L is a (sparse) lower triangular matrix and D a diagonal one; then eq.(5.3) is solved by performing a forward and backward substitution on the L factor. Note that the I_m vector of eq.(5.4) is very sparse itself (single nonzero entry). Therefore to make computations faster, we use the "sparse vector method" of Ref.[15]; this exploits the sparse structure of the LHS vector while performing forward substitution

$$b = Lx. \quad (5.6)$$

The resulting procedure - called "fast forward substitution" (FFS) - identifies which columns of L are involved, i.e. which do not lead to useless zero multiplications. Because there are few such columns, FFS is much faster than the (usual) full forward substitution. Once the solution of (5.6) is obtained, a full backward substitution is performed to obtain that of (5.3).

To summarize, computing Y_o from Y^u requires n fast forward and full backward substitutions.

5.2. Computing the during-fault matrix Y_D

Adapting eq.(5.2) to the short-circuited configuration, we compute Y_D through :

$$[Y_D]_{ij} = I_i]_{E_k=0, k \neq j; E_j=1} \quad (5.7)$$

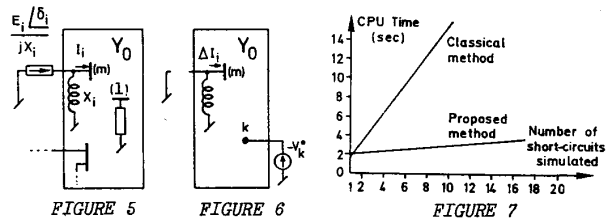
where, by virtue of a well-known superposition technique

$$I_i = I_i + \Delta I_i, \quad (5.8)$$

I_i : the current in the i -th generator in the pre-fault configuration (Fig.5); it is already available from Y_o ; ΔI_i : a correction term obtained as follows : let k be the short-circuited node; let V_{k0} be the voltage at this node in the pre-fault configuration. The effect of the short circuit is obtained by imposing the $-V_{k0}$ voltage to the passified circuit (Fig.6).

To have all LHS currents known in (5.3), we rather impose a unit current at node k and correct $[V_m^T \ V_1^T]^T$ accordingly. The solution of (5.3) for a unit LHS vector has to be obtained only once, for the whole Y_D determination (only the value of V_{k0} changes from one column to another). Moreover the FFS technique can be used again, since a unit vector is involved.

As a consequence, Y_D can be obtained from Y_o very efficiently by performing a single fast forward and full backward substitution on the factorized Y^u .



Remark. In order the various V_{k0} voltages to be available when computing the Y_D matrices, the m vectors $[V_m^T \ V_1^T]^T$ should be memorized in the course of computing Y_o . The full vectors need not be saved; only the components corresponding to nodes (k) to be short-circuited are required.

5.3. Computing the post-fault matrix Y_p

Finally we consider the problem of obtaining any Y_p matrix, differing from the pre-fault one by the outage of a network element (line, transformer, etc.). Eq. (5.3) becomes :

$$\begin{bmatrix} I_m \\ 0 \end{bmatrix} = Y_p^u \begin{bmatrix} V_m \\ V_1 \end{bmatrix} = (Y_o^u - M \Delta_y M^T) \begin{bmatrix} V_m \\ V_1 \end{bmatrix} \quad (5.9)$$

where M is a sparse $(q \times N)$ matrix containing 0s, +1s and -1s and Δ_y is a $(q \times q)$ matrix containing outaged element admittances (generally $q=1$ or 2). The adequate choice of M and Δ_y are further discussed in Ref.[16], from which the following compensation scheme is inspired.

The matrix inversion lemma gives

$$(Y_p^u)^{-1} = (Y_o^u)^{-1} + (Y_o^u)^{-1} M (\Delta_y^{-1} - M^T (Y_o^u)^{-1} M)^{-1} M^T (Y_o^u)^{-1} \quad (5.10)$$

Introducing (5.10) into (5.9) yields

$$\begin{bmatrix} V_m \\ V_1 \end{bmatrix} = \begin{bmatrix} V_{m0} \\ V_{10} \end{bmatrix} + X (\Delta_y^{-1} - M^T X)^{-1} M^T X \begin{bmatrix} V_{m0} \\ V_{10} \end{bmatrix} \quad (5.11)$$

where :

X is the $(q \times N)$ matrix, solution of $Y_o X = M$ and is obtained by performing q fast forward and full backward substitutions on the available factorized Y^u matrix; $\begin{bmatrix} V_{m0} \\ V_{10} \end{bmatrix} = (Y_o^u)^{-1} \begin{bmatrix} I_m \\ 0 \end{bmatrix}$ is the solution obtained in the course of computing Y_o .

Therefore computing Y_p from Y_o amounts to performing q fast forward and full backward substitutions and correcting Y_o accordingly.

5.4. Simulation results

Comparisons have been carried out between the pro-

TABLE II

	Config. #	Fault loc.	CCT by SPS	CCT by direct method	CCT by proposed method
					$\alpha_1 = \alpha_2 = 1$ $\alpha_1 = 0.30$ $\alpha_2 = 1.70$
CIGRE	17	4	49	52	40
	21	3	39	44	35
	21	4	37	39	31
GREEK	01	1	50	56	51
	05	1	51	57	49
	08	1	37	42	54
	09	1	44	53	50
	10	1	41	52	50
	11	1	51	59	54
	11	0	71	73	69

TABLE III

Statistical information on the 27 configurations of CIGRE system

Config.	μ		σ		Config.	μ		σ	
	direct method	new method	direct method	new method		direct method	new method	direct method	new method
0	.001	-.022	.012	.036	15	-.007	-.009	.013	.031
1	-.003	-.018	.009	.027	16	.022	.020	.054	.062
2	-.006	-.008	.012	.028	17	.013	-.090	.033	.158
3	-.004	-.002	.020	.033	18	-.143	.044	.375	.135
4	-.014	-.008	.068	.051	19	.006	-.014	.026	.038
5	-.006	-.024	.018	.037	20	.010	-.038	.056	.043
6	-.008	-.015	.015	.023	21	.035	-.047	.055	.074
7	-.007	-.028	.014	.032	22	.031	.014	.065	.050
8	-.005	-.025	.010	.032	23	.010	-.016	.043	.032
9	-.004	-.019	.011	.025	24	.005	-.018	.035	.025
10	-.014	-.015	.065	.068	25	.006	-.024	.021	.029
11	-.009	-.032	.015	.041	26	.009	-.021	.025	.026
12	-.004	-.020	.010	.034					
13	-.020	-.027	.024	.035	All tests	-.002	-.018	.080	.055
14	-.012	-.023	.018	.038					

Table III provides statistics of CCTs related to disturbances located at all GBs for all 27 CIGRE system's configurations. Similar statistics have been obtained for the Greek system [11]. On the other hand, we report hereafter statistics relative to disturbances applied at both GBs and OBs (42 cases) of the Belgian system; all considered CCTs range up to 0.50 sec.

	μ	σ
GB :	0.00117	0.0506
OB :	0.00660	0.0588

Remark. The optimal couple of values of α_1, α_2 factors differs slightly from one system to another. In practice, its search should be carried out off-line. Particularizing α_1, α_2 values for each system would have resulted in an improvement of the accuracies reported in this section.

6.3. Computing times

We give hereafter a short comparison between the most refined Liapunov based method (global Kyoto approach, accounting for transfer conductances through numerical integration, exact trajectory) and the new method. In both, the technique of Section 5 is used to compute the Y matrices; a separate account, without this calculation is given. A single ($k=1$) as well as a set of ($k=30$) contingencies have been considered. The CPU time of the direct method is considered to be the time required to reach the maximum of potential energy, for a disturbance corresponding to a mean value of CCT (different for each system).

	15-mach.	Greek	Iowa	Belgian
direct method	$k=1$ { without Y	1.36	1.31	1.11
	total	1.73	1.70	3.06
	$k=30$: total	43.78	42.59	41.60
new method	$k=1$ { without Y	0.09	0.07	0.11
	total	0.46	0.46	2.06
	$k=30$: total	5.68	5.39	11.63

As can be seen the proposed method is between 4 and 10 times faster than the direct method of comparable accuracy.

6.4. Salient features

The results given in Tables I, II, III are only a

TABLE IV

Features	Accuracy	CPU time	Other information			Flexibility		Applicability					
			security margin	limit power	voltage etc.	fault location	mode of instability	load or generator shedding	off-line screening	on-line assessment	sensitivity analysis	corrective control	OPF
SBS	benchmark	very important	no	no	yes	both generator and non-generator buses	any type	yes	no	no	too difficult	no	no
DIRECT	good	little	yes	difficult	no			to be investigated	not fast enough	no	possible	possible	no
NEW	good	significantly less	yes	difficult	no			yes	yes	yes	yes	yes	possible

representative sample out of about 1000 simulations reported in Ref.[11]. Table IV summarizes main method's features drawn from both practical and theoretical considerations.

6.5. First implementation results

This method has recently been tested off-line in two practical Chinese systems. These systems are characterized by long distance transmissionlines and weak configurations. The pre-fault configurations and operating points are taken from actual operating data. The first implementation results are quite promising.

7. CONCLUSIONS

A direct method has been developed on the basis of a dynamic equivalent used in conjunction with the equal area criterion and Taylor series expansions. Flexible with respect to the type and location of disturbances, to the sequence of events and the pre-fault and post-fault configurations, and free from numerical integration, it allows one-shot computation of either critical clearing times or stability margins. Further, a general technique has been proposed for computing admittance matrices reduced at the generator nodes. Combining this technique with the direct method results in a simple straightforward approach to on-line multiple contingency evaluation as well as to planning studies. Compared with the most accurate Liapunov criterion it proves equally accurate and faster about one order of magnitude.

Even more appealing is the possibility that the new method offers for on-line sensitivity assessment and control of large power systems. Currently under investigation, the results of this study will be reported soon.

8. REFERENCES

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Discussion

A. Rahimi (Systems-Europe, Brussels, Belgium) and **G. Schaffer** (Brown Boveri, Turgi, Switzerland): We would like to congratulate and thank the authors for this work. It is gratifying and encouraging for us that they have independently followed an approach similar to ours (their Ref. [7]). However, they seem to have overlooked some essential points in [7], that we would like to emphasize for more fruitful pursuit in this area.

1) The authors assume the same rotor angle for all machines comprising an aggregate (cf. Eq. (2.6)). This results in an "incongruent" transformation. In other words, the steady-state equilibrium angle of the resulting OMIB system turns out to be different when computed directly from individual machine steady-state equilibrium angles, than when it is calculated from the OMIB equation. Postfault steady-state consistency between the original and aggregated system representations is a basic requirement, at least from a mathematical point of view, for conclusions regarding correspondence of their dynamic behavior. In [7] the steady-state angle offsets were taken into account in computation of the OMIB system parameters, thus resulting in a "congruent" transformation. Moreover, although the formulas for these parameters were given in [7] only for the simple case involving separation of one machine with respect to the rest of the system, the more general case of two multimachine aggregate separation is treatable in a similar way, as indicated below for completeness and further reference:

Let A and B denote the index sets of the machines belonging to the two aggregates respectively. With no loss of generality assume B is a more massive aggregate. Using the usual notation, define

$$M_A = \sum_{k \in A} M_k \quad (A.1)$$

$$M_B = \sum_{j \in B} M_j \quad (A.2)$$

$$\tilde{M} = \frac{M_A \cdot M_B}{M_A + M_B} \quad (A.3)$$

$$\mu = \frac{M_B - M_A}{M_B + M_A} \quad (A.4)$$

$$\delta_A = \frac{1}{M_A} \sum_{k \in A} M_k \delta_k \quad (A.5)$$

$$\delta_B = \frac{1}{M_B} \sum_{j \in B} M_j \delta_j \quad (A.6)$$

$$\psi = \delta_A - \delta_B \quad (A.7)$$

$$\eta_k = \delta_k - \delta_A; \quad k \in A \quad (A.8)$$

$$\varphi_j = \delta_j - \delta_B; \quad j \in B \quad (A.9)$$

$$b_{kj} = |E_k| |E_j| B_{kj} \quad (A.10)$$

$$g_{kj} = |E_k| |E_j| G_{kj} \quad (A.11)$$

Assuming constant turbine powers P_{mi} , and constant internal machine emf's, the following equation is obtained by purely mathematical manipulation of individual machine dynamical equations without any additional assumptions:

$$\tilde{M} \ddot{\psi} = \tilde{P} - P_{\max} \sin(\psi - \nu) \quad (A.12)$$

where

$$\tilde{P} = P_m - P_c \quad (A.13)$$

with

$$P_m = \tilde{M} \left(\frac{1}{M_A} \sum_{k \in A} P_{mk} - \frac{1}{M_B} \sum_{j \in B} P_{mj} \right) \quad (A.14)$$

$$P_c = \tilde{M} \left[\frac{1}{M_A} \sum_{k,i \in A} g_{ki} \cos(\eta_k - \eta_i) - \frac{1}{M_B} \sum_{j,i \in B} g_{ji} \cos(\varphi_j - \varphi_i) \right] \quad (A.15)$$

and

$$P_{\max} = \sqrt{\left\{ \sum_{k \in A} \sum_{j \in B} [b_{kj} \cos(\varphi_j - \eta_k) + \mu g_{kj} \sin(\varphi_j - \eta_k)] \right\}^2 + \left\{ \sum_{k \in A} \sum_{j \in B} [b_{kj} \sin(\varphi_j - \eta_k) - \mu g_{kj} \cos(\varphi_j - \eta_k)] \right\}^2} \quad (A.16)$$

$$\nu = \tan^{-1} \frac{\sum_{k \in A} \sum_{j \in B} [b_{kj} \sin(\varphi_j - \eta_k) - \mu g_{kj} \cos(\varphi_j - \eta_k)]}{\sum_{k \in A} \sum_{j \in B} [b_{kj} \cos(\varphi_j - \eta_k) + \mu g_{kj} \sin(\varphi_j - \eta_k)]} \quad (A.17)$$

The parameters P_c (and thus $\tilde{P}$), $P_{\max}$, and ν so defined are time-varying. They change with changes in network topology (the b 's and g 's) as well as machine angle variations in each aggregate (η 's and φ 's). Using postfault topology and steady-state angles to compute the postfault OMIB parameters yields a congruent aggregation in the sense defined above. The choice of steady-state values is further justified by the fact that during the postfault system evolution, the variations of offset angles in each aggregate (and thus the time varying OMIB parameters) are almost periodic around their postfault steady-state values. The authors of the present paper have in fact arbitrarily set all η 's and φ 's to zero by virtue of their assumption (2.6).

2) The authors use the aggregated OMIB system equation to find critical clearing times (CCT) directly. In [7] we introduced and used a similar concept for the case of "Constant Power Loads." However, in the case of "Constant Impedance Loads," although we had explored this possibility (cf. Section V.B of [7]), we decided not to use it due to its potential pitfalls. The point is that during the fault-on period, the changes in angle offsets of machines comprising an aggregate are not "almost periodic," but vary almost with a "polynomial in t ." If the fault duration is short, the machine angle offsets will not change much during the fault-on period; thus one may use their prefault steady-state values (along with fault-on values for the b 's and the g 's) to compute the fault-on OMIB parameters. However, for long faults, neither the prefault nor the postfault steady-state values of angle offsets are relevant. The erratic behavior of the OMIB parameters during some long fault-on periods, led us in [7] to use "leap-frog" integration on individual machines instead. This precludes direct computation of CCT. Having ignored OMIB parameter dependence on machine angles, the authors of the present paper use "leap-frog" integration on the OMIB system directly and rely on empirical correction factors. Unfortunately, there does not seem to be a clear mathematical justification for the choice of such factors; the authors only refer to a "learning simulations set" they have experimented with. We would like to reiterate the need for caution in this area. The authors do report cases where without such correction factors the method would not have worked. It would be interesting to study the variation of the exact OMIB parameters during fault-on conditions for these cases, using standard simulation programs and equations (A.13) to (A.17) to assess the error introduced by assumption (2.6).

3) As the authors point out in Section 3.4.1, the notion of a CCT is pertinent only to single-action faults. The concept of a stability index is more general and covers faults with successive actions as well. If properly defined, a stability index could be used to identify the critical machines. This was explained in Closure of Ref. [7]. The authors have proposed the use of CCT for the same purpose (cf. Section 3.3). We do agree that for the cases where CCT is pertinent and safely computable it is a good measure to use; but for more general cases the stability index is the only possibility.

4) To encourage the pursuit and use of the approach presented in this work and Ref. [7], we would like to mention its amenability to parallel processing. Stability indexes for various machines (or candidate aggregates) can be computed in parallel, rather than sequentially, once the network computations are done. This will further reduce the already small computation time needed, and makes the approach even more attractive for on-line stability assessment in large systems.

We would like to once again commend the authors for this work.

Manuscript received February 20, 1987.

Y. Tamura (Waseda University, Tokyo, Japan) and **H. Sasaki** (Hiroshima University, Higashi-Hiroshima, Japan): The authors are to be commended for their fine work and very interesting approaches for fast transient stability assessment of multimachine power systems based on a clever combination of i) the identification of a (cluster of) critical machine(s), ii) the network reduction, and iii) the application of the Equal-Area-Criterion to a reduced equivalent network.

The authors claim that the proposed method is particularly suitable to on-line dynamic security assessment. Although the present discussers well recognize the importance of developing such an approach, there are still several points to be clarified further. We would like to have the authors'

comments on the following points.

El-Abiad and K. Nagappan first applied the Lyapunov's direct method to transient stability analysis of a multimachine power system in 1967. The main advantages of the direct method were the considerable curtailment of solution time and ease in judging stability. Another salient feature to be emphasized is the sufficiency of its results. In the discussers' opinion, this characteristic is the most favorable property when applied to on-line security analyses. It seems that this favorable property has sometimes been overlooked in recent years. When we talk about the critical clearing time (CCT) in evaluating a fast stability assessment method such as proposed in this paper in the environment of simulation studies, there would be no problem. A quite different interpretation and critical view would be needed, when such a method is implemented in the energy management software of some utility to be used on-line. Let us notice, for instance, configuration #8 of the Greek system in Table II. The CCT obtained by the step-by-step, the direct, and the proposed methods are, respectively, 37, 42, and 54. Assuming that the proposed method be used in an on-line environment, how would a system operator react on having only the result of "54" for this particular case. It is natural to assume that no reference CCT value by the step-by-step method is available in this circumstance. If the operator regards the system viable for an assumed fault, and if the anticipated fault would actually occur, what would happen on this system? Recent trends which look to pursue how to improve the ratio of hitting the mark with less computation time without the explicit proof of sufficiency may run up against a wall.

In developing the proposed approach, the authors assume that all generators in a studied system be separated into two groups. According to our experiences, however, generators may be grouped into more than two clusters depending on the location of a fault. How would the proposed method cope with such cases?

Assumption (2.6) seems to play a fairly important role in deriving the new scheme. Would the authors be able to show good evidence to justify this assumption?

Presumably, the most bewildering aspect of any direct method is in that it gives good results for some fault cases while it produces rather poor consequences for other contingencies. According to our findings, good correspondence is generally obtained for a fault assumed at a point very near a generator bus. This statement may well be fortified by the results in Table I in the paper.

The authors are cordially requested to give their views on the points mentioned above. Again, we would like to commend the authors for their fine work.

Manuscript received February 25, 1987.

G. A. Maria and M. Zein El-Din (Ontario Hydro, Toronto, Ontario, Canada): The authors present a useful technique for the reduction of large power systems to two machines suitable for the application of the equal area criteria. This (or similar) technique will be useful for the on-line application of the energy method to obtain approximate transfer limits across critical interfaces. We have a number of comments and questions:

1) One machine is used to represent the critical machine(s) and another is used to represent the remaining machines. How good is this approximation when dealing with a large number of critical machines separating from a large interconnected system? Also, the technique presented is based on "classical machine" representation; is there a feel for how conservative this model is in on-line applications? Further, could generator excitation control effects be incorporated in the proposed techniques, and if so, how would that alter the process of identifying the critical machine(s).

2) It is, in general, assumed that a faulted system will split along its weakest mode. It is also likely that machines with large post-fault speed and/or acceleration would be in the critical machines list. The authors are right in proposing the acceleration criteria for the identification of critical machines candidates. What about the speed component? Different ways can be used to identify the faulted system weakest mode. In this paper, the weakest mode is identified as the one with the smallest critical clearing time. In [1], the weakest mode is identified as the one with the smallest ratio between the potential energy that the system can absorb and the kinetic energy at fault clearing. It is computationally a little harder to identify the critical machines using the critical clearing time criteria. What are the advantages of the proposed technique?

3) When dealing with multiple contingencies, the proposed Y-matrix reduction will have a definite advantage over the techniques designed for single contingencies. However, the advantages of the proposed technique are overstated in the paper. If the buses associated with faults and equipment switching are moved to the bottom of the matrix, one has to perform full reduction only once for the prefault conditions. Partial

reductions are used for fault and postfault conditions. The component (d) in the computation time equation is about two times (b). The computational advantage of the proposed technique is one-to-two times depending on the number of contingencies, not the order of magnitude reported.

4) The proposed Y-matrix reduction technique handles the fault Y-matrix and the postfault Y-matrix reductions differently. This requires the development of different software. The difference between the fault Y-matrix and the prefault Y-matrix can be expressed as $(M\Delta Y M^T)$. The proposed procedure for the postfault Y-matrix reduction can be used to produce the fault Y-matrix reduction.

5) The proposed Y-matrix reduction technique requires the saving of $[V_m V_i]$ arrays. For a system with 1000 buses and 100 generators, this represents 100k word storage. This storage requirement can be easily eliminated by modifying equation (5.11) to:

$$Y_p = Y_0 + Z(\Delta Y^{-1} - M^T X)^{-1} Z^T$$

where

$$Z = [I_M] X$$

and the rows of I_M are the I_m arrays of equation (5.4).

6) Now that the arrays $[V_m V_i]$ are no longer required to compute the fault and postfault Y-matrix reductions, the normal reduction technique can be used to compute Y_0 and the factor $[LDL^T]$. This will result in some computation saving ($a > c$ in the computation time equation).

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Manuscript received February 25, 1987.

B. C. Papadakis and A. V. Machias (National Technical University of Athens, Greece): The authors are to be congratulated for their valuable work. Their effort to update the direct method of equal-area criterion for fast transient stability assessment suitable for on-line application is quite laudable and the cases studied in the paper to illustrate the proposed method excellent.

Indeed, the well-known equal-area criterion based on the equivalent two-machine model with the classical system simplifications and assumptions is probably the most suitable methodology to obtain fast estimation of transient stability, gifted with the following advantages. It recognizes type and location of disturbance, sequence of events, and stability limits and margins. Moreover, the method can handle disturbances other than faults. However, the main drawback of this approach is the simplicity of the two-machine (or, one-machine to infinite bus) equivalent which can only approximate an actual multimachine, multibus interconnected electric power system, with lossy components, nonlinearities, high-speed exciters, controllers and stabilizers, regulators, and reactive compensators, having an overall considerable contribution to the transient and dynamic behavior of the system. Since the above contribution is normally beneficial to system stability, the direct methods can only claim a conservative estimation, although valuable in on-line mode.

With the above conditions, any additional simplification introduced in the analysis, or in the computational procedure will further weaken the overall accuracy of the results. To this aspect, we would comment on the assumption (2.6) of the paper, equalizing rotor angles within the set of remaining $(n - 1)$ machines. In a physical sense this assumption distorts somehow the profile of power flows in the system, modifying accordingly the electric output of the specific machine. In the analytical model it affects the elements of the Y_0 matrix, as can be seen from (5.2) and Fig. 5, and subsequently the elements of Y_D , Y_p . We suppose that abandoning the assumption would result in somewhat longer cpu's in favor of the accuracy of results. Have the authors evaluated this effect?

Apart from the examined disturbances on the generator buses, it is also important in many cases to investigate the effect on stability of faults in the central buses of the system. Such cases can be found in the Greek system which is comprised in the studied cases of the paper. This computation takes more time due to additional matrix reductions. The authors report faults on buses other than generator ones (OB), but it is not very clear if they comprise them in their results.

Again the authors are complimented for their paper.

Manuscript received February 26, 1987.

S. S. Venkata (Universität Duisburg, West Germany and University of Washington): The authors are to be congratulated for proposing a simple, computationally fast, and practical method for evaluating the transient stability of large power systems. The paper is very clearly written incorporating several novel techniques such as the sparse vector method for computing the prefault admittance matrix, the compensation method for determining the during-fault matrix, and the Sherman-Morrison algorithm for finding the postfault matrix of the system. These features will undoubtedly speed up the computation process required for on-line applications. Above all, the real power of the method lies in the fact that the entire formulation is algebraic in nature.

I have, however, the following questions:

1) Why did the authors prefer η as the stability margin for the sensitivity analysis? I believe that CCT is a better performance index for the sensitivity analysis, since it can provide additional information to an engineer for the optimum design of protection schemes for the critical machines.

2) How can one translate and relate the results of the reduced system to the real system? In other words, is there any inverse procedure or transformation available, to proceed from the one-machine infinite bus model to the multimachine model, to relate the results obtained to each and every critical machine individually?

3) Can the proposed method be used for real-time applications?

4) A philosophical question: Are Liapunov methods, as applied to power systems, obsolete?

In my opinion, the paper is a valuable contribution to the literature and I strongly encourage the authors to extend the work and continue to report their further findings.

Manuscript received March 3, 1987.

Sheng Liu (Shanghai Jiaotong University, Shanghai, China): The authors are to be congratulated on devising a simple but effective method for fast transient stability assessment of large power systems. We believe in the authors' effort that the proposed method is about one order of magnitude faster than the current direct method of comparable accuracy. Therefore, it is useful either for on-line contingency analysis or off-line planning studies with its speed advantage.

The authors' clarification is sought in the following areas:

1) Identifying the critical machine for various disturbances is a very important procedure for the direct method [1]. The authors use the initial acceleration criterion [2] to "pre-filter" a list of "candidate" critical machines. Since it does not identify the critical machine, how to draw up a list of "candidate" critical machines is still a problem. In other words, some simple rules for determining the "candidate" critical machines are desirable. Could the authors comment on this?

2) The authors' comments on the effect of the proposed method subjected to unsymmetrical faults will be appreciated.

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Manuscript received April 3, 1987.

Y. Xue, Th. Van Cutsem, and M. Ribbens-Pavella: We are thankful to the discussers for their valuable and constructive comments. In order to answer their questions in detail, it is useful to first clarify some aspects of our method to which we henceforth refer to as the "Extended Equal Area Criterion" (EEAC). The EEAC essentially pursues two objectives: i) Devise an ultrafast technique to assess transient stability (in terms of CCT's or margins as appropriate); this goal was met in this paper. ii) Provide analytical means of sensitivity analysis and control; initiated here, this study is developed in Ref. [A]. To reach these objectives, the EEAC method has called upon one conjecture and one simplifying assumption. It is worth restating them and examining their implications in some detail.

Conjecture—The disturbed system's separation depends upon the maximum angular deviation between the following two—and only these two—equivalent subsystems: the critical one, generally composed of a single machine, and the remaining (i.e., most of the system) machines; let,

as in the paper, δ_c and δ_a denote the rotor angles of the corresponding aggregated critical and COA subsystems.

Physically speaking, this conjecture seems quite sound, at least within the simplified individual machine model used here. Indeed, it is quite reasonable to consider that the responsible for instability is the negative synchronizing torque appearing between the above two equivalent machines, even if within the COA group many subgroups may actually be formed, which coherently—but more slowly than the critical—are swinging with respect to each other. This conjecture amounts to identifying the CCT as being that clearing time for which the angular deviation $\delta_c - \delta_a$ reaches about 180° . We believe that, irrespective of the method to which it applies (SBS or EEAC), this criterion is more sound than the conventional one, according to which instability is reached as soon as the largest rotor angular deviation among *any* two machines reaches 180° . Note incidentally that by essence it is more sound to rely on the COA devised here, rather than on the "global COA" which incorporates the critical machine (or cluster).

According to our experience, the above conjecture proves to be always reliable for the identification of the actual CCT, whereas the conventional criterion fails in a significant number of cases, of which many are perfectly realistic.

As an illustration, let us report some simulation results concerning a (the "18th") modified configuration of the CIGRE system, subject to a 30SC at GB #5. The computations have been carried out by SBS. According to the 180° criterion applied to the maximum rotor angle deviation between any two machines, the critical clearing time is declared to be 0.54 s whereas the actual one is 0.60 s. Figures A and B plot the swing curves for the fault

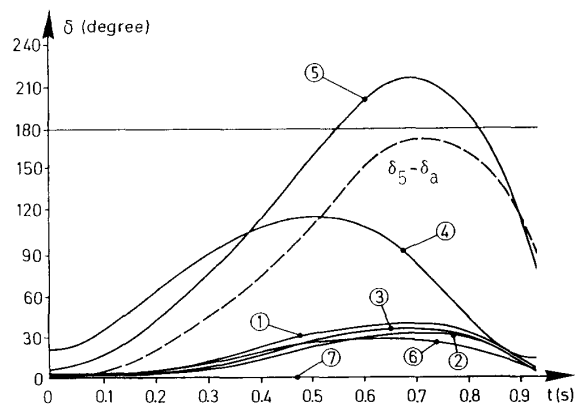


Fig. A

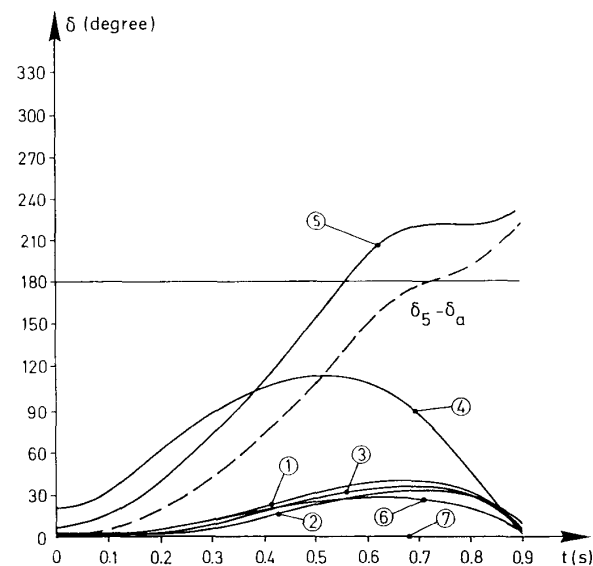


Fig. B

cleared at 0.59 s and 0.60 s, respectively. One can see that for 0.59 s, $\delta_5 = \delta_7$ goes past 180° (at 0.55 s), reaches 217° (at 0.68 s), then decreases and remains below 180° ; on the contrary, for 0.60 s, synchronism is definitively lost. The criterion based on the $\delta_5 - \delta_7$ angular difference identifies unambiguously the correct CCT.

Assumption—Within the aggregated COA subgroup, the rotor angles of the individual machines, $\delta_i(t)$, $\forall i, j \in A$, are supposed to behave according to the following twofold assumption. A1: they are “frozen” with respect to $\delta_a(t)$, and hence $\delta_{ij}(t) = \delta_i(t) - \delta_j(t)$ is constant throughout the fault-on and postfault periods; A2: they moreover assume the same value, $\delta_a(t)$, and hence $\delta_{ij}(t) = 0$. A1 proves to be vital for the elaboration of the very EEAC method; indeed it is only when A1 holds that the δ_{ij} 's are time invariant; this is a necessary condition for the parameters P_c , P_{max} , ν to be constant. Unfortunately, this assumption unavoidably introduces some inconsistencies discussed hereafter.

Assumption A2 on the other hand may *a priori* be thought of as just one way to realize δ_{ij} 's constancy. Other means are also available. Let us quote possible alternatives and their implications.

i) $\delta_i = \delta_a \forall i \in A$, $\delta_{ij} = 0$: the OMIB solutions δ_o (and δ_p) will be different from the prefault (and the postfault) load flow solutions;

ii) $\delta_i - \delta_j = \text{constant} \neq 0$: if

- $\delta_{ij} = \delta_{ij}|_{t=0}$, then δ_o and the prefault load flow solution become consistent but δ_p is different from the postfault load flow one;
- $\delta_{ij} = \delta_{ij}|_{\text{postfault solution}}$, then the inconsistency applies to δ_o and the prefault load flow solution.

To choose among the various possibilities, let us bring out an additional observation. It concerns the computation of δ_a during the fault-on period via Taylor's expansions (here τ denotes a clearing time). With the conditions of the above case i), one can easily see that it is indifferent to compute the equivalent aggregated angle either by means of the COA's evolution with τ or by means of the individual machines' evolution. Indeed expanding δ_a of COA yields

$$\delta_a(\tau) = \delta_a(0) + \frac{1}{2} \gamma_a \tau^2 + \frac{1}{24} \gamma_a^{(2)} \tau^4 + \dots \quad (\text{C.1})$$

On the other hand, if we expand the individual rotor angle $\delta_i(t \in A)$

$$\delta_i(\tau) = \delta_i(0) + \frac{1}{2} \gamma_i \tau^2 + \frac{1}{24} \gamma_i^{(2)} \tau^4 + \dots \quad (\text{C.2})$$

then we sum up all the terms and apply (2.3), we get again (C.1). On the contrary, this is not possible in cases ii). More precisely, what is not possible is to assume that the δ_{ij} 's remain time invariant and that they expand according to a Taylor's series. Indeed, expanding δ_j by an expression similar to (C.2) and considering δ_{ij} yields

$$\delta_{ij}(\tau) = \delta_i(\tau) - \delta_j(\tau) = \delta_{ij}(0) + \frac{1}{2} \tau^2 (\gamma_i - \gamma_j) + \frac{1}{24} \tau^4 (\gamma_i^{(2)} - \gamma_j^{(2)}) + \dots$$

Hence, $\delta_{ij}(\tau)$ cannot be time invariant unless all γ 's and their derivatives vanish. In conclusion, only the A2 assumption makes the OMIB representation totally equivalent to the aggregation of the individual rotor angles, i.e., makes licit the direct integration on the OMIB system.

The above analysis is not exhaustive. Many corrections could be tentatively proposed. Such a simple, yet efficient, correction could be inspired by a practice currently used within the context of the Liapunov direct criterion; it would consist in recomputing an equivalent mechanical output power, P'_m , so as to make the postfault load flow solution coincide with δ_p given by the OMIB. Figure 4 shows that P_m changes can be easily handled. This improvement is certainly worth exploring. But even without it the method yields satisfactory results in terms of CCT. Of no lesser interest is the advantage to make available a simple analytic expression of the transient stability margin and therefrom to provide effective tools for on-line sensitivity analysis and preventive control.

A) Drs. A. Rahimi and G. Schaffer

1) From the foregoing discussion, it is obvious that the assumption A2 results in an “inconsistent” or “incongruent” transformation. Note that if from a mathematical viewpoint this difficulty is indeed awkward, from a practical viewpoint it generally does not influence significantly the accuracy. (With respect to the latter aspect please see our answer to Profs. Tamura and Sasaki, to follow.)

Your expressions (A.1) to (A.17) are also in agreement with the previous discussion. Unfortunately, the time varying expressions of P_c , P_{max} , and ν cannot be used within the approach of [7] either except for the very particular cases where i) $k = 1$, or when many machines are connected at the same busbar; ii) the 30SC is located at this busbar; iii) the critical machine (group) is k . Otherwise, the assumption used in [7] ($P_{max D} = 0$) is

not valid. (Incidentally a typing error in (A.14) transformed $\bar{M}^{-1}$ and $\bar{M}_A^{-1}$ into $\bar{M}$ and $\bar{M}_A$, respectively.)

2) Your discussion about the physical plausibility of the various assumptions is very interesting, indeed. Unfortunately, we cannot take advantage of it, since our approach is bound to use A2.

With regards to load modeling, the EEAC can consider constant impedance loads only, at least for the time being.

Concerning the corrective factors α_1 , α_2 , admittedly the justification given in the paper does not have mathematically rigorous foundations, since these factors are introduced to diminish to the extent possible truncation errors of a Taylor's expansion, which is also an approximation. The idea is to apply this series to small time intervals (smaller than the actual CCT), by successively reducing the clearing angle scale—applying Taylor series—amplifying the time scale. The procedure is described later in this closure, in answer to a question of Drs. Maria and Zein El-Din. It is realized by using the factors $\alpha_1 < 1$, $\alpha_2 > 1$, which moreover obey the constraint (3.11) shown in the paper. Observe that the accuracy is scarcely affected by α_1 , whereas it is more sensitive to α_2 ; this follows from the more linear character of the curves of Fig. 2 in the range of small values of clearing times than of large ones. Observe also that the values can easily be adjusted and that the optimum α_2 values may differ from one to another power system, and even from one to another generator bus. Taking advantage of this fact would probably have improved the already good accuracy of the simulation results reported in Section 5. This adjustment can easily be performed off-line for a given power system, and this is the meaning behind the somewhat esoteric term of “learning simulations set.”

3) We believe that the CCT's and stability margins are two different means of TSA, each with interest on its own. But by definition, the identification of the critical machine(s) should rely on the criterion of the minimum clearing time rather than that of the margin. An easy way to illustrate that the latter criterion may fail to identify the critical machine(s) is to use the equivalent notion of Liapunov function and its limit value, supposed to be estimated correctly. Figures C and D show respectively a

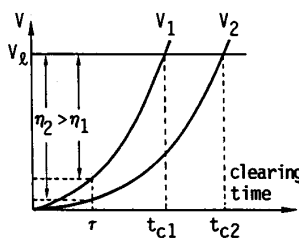


Fig. C

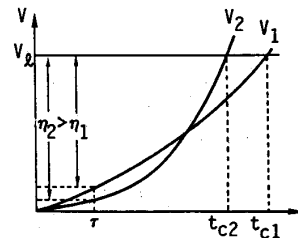


Fig. D

more or less “normal” situation: one can see that in Fig. C, η and CCT provide the same information, whereas in Fig. D they are in contradiction.

4) We agree that parallel processing schemes can certainly speed up the procedure. We think, however, that the already short computing time of the stability index calculation *per se* is not worth a parallel scheme. It looks more attractive to us to process the reduced admittance matrices computations in parallel. Within the context of compensation methods, it appears appropriate to treat one contingency on one processor. One could imagine for instance a “master” being in charge of computing Y_0 and sending the relevant terms to “slaves.”

B) Professors Y. Tamura and H. Sasaki

We agree that at its beginning, the application of the Liapunov criterion to transient stability was used in conjunction with the “theoretical stability domain estimate” which is recognized to provide conservative results; moreover, the degree of conservativeness varies in a large range of values, and in a random fashion with the system and with the fault type and location. This was making the method of little, if any, practical interest. Nowadays, however, by means of “practical stability domain estimates” the Liapunov criterion has become totally reliable and quite accurate. Such an interesting approach was proposed by Kakimoto and Hayashi [B]; together with a very simple practical safeguard [4], it provides necessary and sufficient practical stability assessment.

Concerning the EEAC accuracy, we have indeed found and reported a few cases with large errors. Note, however, that: i) unacceptable discrepancies are very few; ii) they are likely to be easily identifiable; this is suggested by the kind of approximations implied and corroborated by our experience; iii) the “anomalous case” identification may be performed off-line, then taken into account on-line to set aside all unreliable information.

Observe also that the EEAC is up to a certain extent still perfectible; a couple of possible, easy improvements have been suggested above.

As regards your question relative to the case where the generators are split into more than two groups, we lack experience. We could, however, tentatively suggest to use a particular energy Liapunov function, constructed with similar to our basic assumption (2.6), and comprising as many equivalent machines as there are coherent groups.

C) Drs. G. A. Maria and M. Zein El-Din

1) It would be interesting, indeed, to explore the behavior of the EEAC in the case you mention of a very large system with a large number of critical machines. The validity of the classical machine representation is also a good still-open question; no consensus seems to exist. Refining machine and load representation is of course always desirable. We believe, however, that increasing the method's complexity above certain limits could make it depart from the essential objectives for which it has been designed.

2) To identify candidate critical machines we use the initial acceleration criterion—and this indeed does not account for speed components. But the final critical machine(s) identification relies upon the minimum among the CCT's of the above candidates; and within the model limitations, this contains a good deal of information, including speed at the clearing time.

The concept of weakest mode proposed in your Ref. [1] is certainly very interesting. The margin you propose for its identification is

$$\Delta V = \frac{V_l - V_p(\tau)}{V_k(\tau)} \quad (C.3)$$

where $V_p(\tau)$ and $V_k(\tau)$ are the potential and kinetic energies of $V = V_p + V_k$, and where V_l stands for the limit value of V . This is an alternative to more conventional margins such as the expression $(V_l - V)/V_l$, or the expression $(V_l - V)$ sketched in Figs. C and D. Like this latter, the margin (C.3) may result in some misclassifications, as the one illustrated in Fig. F where the smaller margin corresponds to a more stable case than the larger

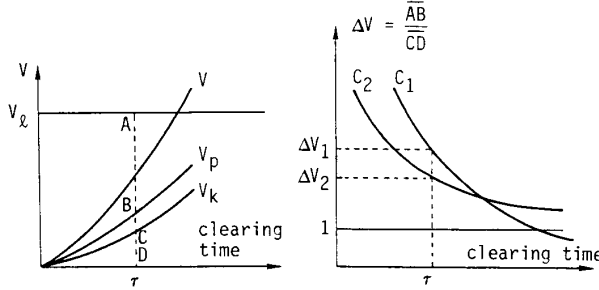


Fig. E

Fig. F

one. Note that in general, the use of a Liapunov function in the multimachine case makes difficult the derivation of analytically tractable expressions for margins.

Your observation according to which it is computationally a little harder to identify the critical machines using the critical clearing time criterion, is indeed valid for the Liapunov function-based methods. For the EEAC, however, this does not hold. To clarify our statement, let us compare the specific steps involved by CCT and η , as they are brought out in Section 3.2.

- To get the margin $\eta(\tau)$: i) compute $\delta_r = f(\tau)$ from (3.2); ii) compute $A_{acc}(\tau)$ and $A_{dec}(\tau)$ via eqs. (3.4), where, however, δ_r is used instead of δ_c ; iii) apply one of the expressions (3.14) as appropriate.
- To get the CCT $\equiv t_c$: i) compute δ_r from (3.5); ii) express δ_r from (3.9); iii) compute the τ corresponding to δ_r from (3.7); iv) derive t_c from (3.10).

Obviously, the computing times of the above steps are extremely short and almost totally negligible with respect to that required by the computation of the Y matrices; this remains true even when the procedures are used repeatedly.

3 to 6) We thank the discussers for their valuable comments on the Y -matrix reduction techniques; they may contribute to enhance the proposed method's computational efficiency.

The reduction method labeled "classical" in the paper is a "brute force" one; it was chosen as the reference because it is a common calculation procedure, and hence allows easy comparisons. With respect to this method, the gain in computing time is not overstated. Now it is clear that this classical method can also be made more efficient within the context of multiple contingency analysis. In this respect, we do not consider node

renumbering as the most attractive solution, since i) it is not necessarily possible to have ALL the buses involved in network changes at the bottom of the matrix; ii) this reordering may increase the fill-in effect during factorization. An attractive method is the partial matrix factorization as described in, e.g., Ref. [C]. The question is whether either compensation or partial refactorization is best suited to the network changes considered here. The fact remains that imposing the machine internal nodes to be numbered last, disturbs the optimal ordering scheme as mentioned above. In addition, performing partial refactorizations for numerous noncumulative network changes imposes an extra computational overhead which should not be underestimated. The one-to-two ratio experienced by the discussers is very dependent on the network under concern. Similarly, the fact that $a > c$ (as in our simulation results) does not necessarily hold in any case: it depends on the ratio "number of machines/number of nodes," among others.

The proposed modification of (5.11) yields indeed a more elegant and less memory consuming procedure. In fact, (5.11) should be read:

$$\begin{bmatrix} V_m \\ V_l \end{bmatrix} = \begin{bmatrix} V_{m0} \\ V_{l0} \end{bmatrix} + X(\Delta_y^{-1} - M^T X)^{-1} M^T Y_0^{-1} \begin{bmatrix} I_m \\ 0 \end{bmatrix}$$

and the discussers' improvement should be read:

$$\begin{bmatrix} V_m \\ V_l \end{bmatrix} = \begin{bmatrix} V_{m0} \\ V_{l0} \end{bmatrix} + X(\Delta_y^{-1} - M^T X)^{-1} X^T \begin{bmatrix} I_m \\ 0 \end{bmatrix}$$

The compensation method can indeed be used for the faulted configuration also. The most efficient correction consists in adding an adequate admittance to the faulted bus, which requires an $N \times 1$ array of the type M and hence a single fast substitution as in the method of the paper. The "software" is not significantly "different."

As regards the overall method's efficiency we would like to stress the following: i) unfortunately, the discussers' interesting suggestions do not improve significantly the computing times (concerning the potential improvement $a - c$, please refer to the above remark); ii) anyhow, the CPU time is small and will at worst remain as such; iii) the real gain in CPU time to be considered is the ratio:

$$\frac{\text{best time for reduction + time for direct method}}{\text{best time for reduction + time for the EEAC method}}$$

This ratio has not been overstated in the paper and the conclusions of Section 6.3 are fully valid.

Finally, let us make two additional comments:

- the larger the number of generators, the more computationally interesting the EEAC with respect to other direct methods;
- this gain is even more important for cases where the postfault configuration is different from the pre-fault one.

D) Professors Papadias and Machias

Admittedly, the EEAC introduces many simplifying assumptions, more than the conventional Liapunov energy type criterion. This does not necessarily imply deterioration of the accuracy of the EEAC with respect to that of Liapunov's criterion. Indeed, the former assumptions do not necessarily include all the latter and additional ones. Besides, the simplicity of the EEAC approach suggests efficient, easy to master, corrective factors which allow to enhance significantly its reliability. Nevertheless, the Liapunov criterion remains more accurate, and in fact very accurate, indeed.

With regard to the accuracy of the Y -matrix computations, notice that the assumption (2.6) affects only the expressions of P_c , P_{max} , and ν . This has no influence on the matrices Y_o , Y_D and Y_P , nor does it affect their computation. With regard to computing times, there is no difference between these matrices' computations for OB's or GB's cases. But the accuracy of the simulation results is slightly better for GB's than for OB's as it is brought out by the comparisons given, in the paper, below Table III.

E) Professor Liu

The acceleration criterion proves to be a quite appropriate "prefilter" for drawing up the list of candidate critical machines. Indeed, the initial machines' accelerations contain essential information for the type of disturbances currently considered in transient stability studies. Their computation on the other hand does not require any effort additional to that implied by the very EEAC method.

Unsymmetrical faults can be treated within the EEAC in a rather conventional way. It amounts to imposing in the direct sequence network a nonzero equivalent impedance at the faulted bus; this impedance is obtained by adequately combining the Thevenin impedances of the inverse and zero sequence networks as seen from the faulted bus. The use of the compensation method is also straightforward.

F) Professor S. S. Venkata

1) The reason why we prefer η_2 , despite the nice dimensionless property of η_1 , is that for sensitivity analysis purposes we are led to consider partial derivatives of η with respect to important system parameters. Much more tractable algebraic expressions may be obtained with a polynomial rather than a rational function.

We certainly share your opinion that the notion of CCT provides multi-aspect, sound information. Moreover, in the method we propose, CCT is the prerequisite to any subsequent computation. Now, the notion of margin opens a variety of possibilities, e.g., the easy computation of transient stability constraints. It is a kind of dual TSA aspect.

2) The answer to this very thoughtful question is that there is no inverse procedure from the OMIB to the multimachine system. For certain applications one might, however, use approximate substitutes, e.g., participation factors, in order to share and assess limit power flows on important tie-lines. This matter is still far from maturity.

3) Yes, indeed. In our opinion, there is a twofold possibility: i) use the method *per se* for an on-line screening, along with sensitivity coefficients derived in [A]; ii) use it as an analytical tool able to provide heuristics and attributes which could be inserted in a more comprehensive, artificial expert-like approach to transient stability analysis and control.

4) You are certainly aware of the fact that Liapunov's methods are not

obsolete for various reasons. To quote only a few: i) with respect to time-domain solution they are much faster and more flexible; ii) with respect to EEAC, they are more accurate; iii) they have real capabilities of extensions and refinements; iv) they are based on notions essential to physical or engineering understanding—and that of course is very important. After all, the EAC, EEAC, and margins can be considered as by-products of the Liapunov direct method.

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